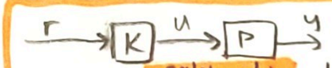
ME 132 Cheat Sheet MT 1

Intro: basis of all feedback sys: ① sense → ② compute → ③ actuate.

Open vs Closed loop.



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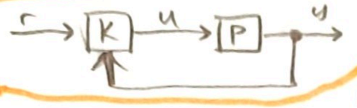


calibration heavy

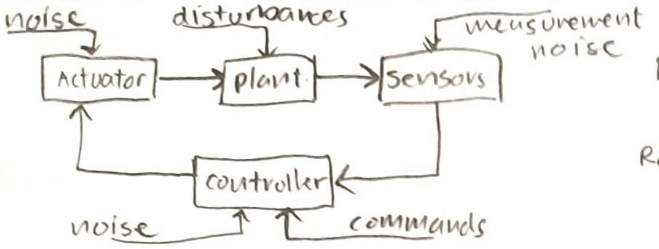
Models: approximations of physical reality! NO MODEL IS PERFECT!

- modelling errors → capture difference between models and physical reality. Ex. ±5% error

vocab

- plant: physical system to-be controlled (creates limits to what controls can do).
- controller (K): brain
- control (u): knob position
- measurement (y): temperature
- reference (r): target temp

STRUCTURE OF CONTROL SYSTEMS



Performance Tracking Metrics ↗ depends on application

- tracking error - settling time - min fuel

Disturbance rejection: do random noise get handled well.

Robustness: Does plant model P_0 work on true plant P?

CAR example

$m\ddot{y} = cu - b\dot{y}$ // dynamic version.

$y = Gu - Hd$ // static car model SIMPLER!

- y = speed
- r = desired speed
- u = throttle motor voltage
- d = percent hill grade
- G = mph/volt (constant)
- H = mph/% (constant)

Nominally.. $G^\circ = 20$ $H^\circ = 5$ nominal model $y = G^\circ u - H^\circ d$

d → is a disturbance!

Static modelling errors: # of ppl in car, throttle effectiveness (G value)

Dynamic modelling errors: wind resistance, rolling friction, delay in fuel lines.

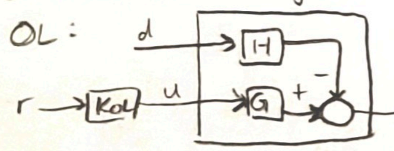
Design objectives

① Perfect Tracking → $y = r$ when $d=0$, $G=G^\circ$, $H=H^\circ$

② Disturbance rejec → $y \approx r$ when $|d|$ is small, $G=G^\circ$, $H=H^\circ$

③ Robustness → $y \approx r$ when $|d|$ small, $|G-G^\circ|$, $|H-H^\circ|$ also both small.

OL:



① $y = G^\circ u + H^\circ d$

$u = K_{OL} r$

$y = G^\circ(K_{OL} r) + H^\circ d$

when $d=0$

$y = G^\circ(K_{OL} r) = r$

$\boxed{K_{OL} = \frac{1}{G^\circ}}$

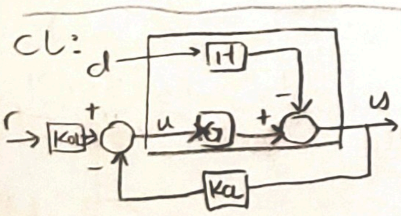
// required for perf. tracking

② $y = G^\circ u + H^\circ d$ { when d=1%, Δy = 5mph

this is ASS

③ $y = \frac{G}{G^\circ} u + Hd$ when $G/G^\circ = 1.1$, $y = 1.1r$ this is ASS

CL:



① $y = G^\circ u + H^\circ d$

$u = K_{OL} r - K_{CL} y$

$y = G^\circ(K_{OL} r - K_{CL} y) + H^\circ d \to 0$

$y = \boxed{\frac{G^\circ K_{OL}}{1 + G^\circ K_{CL}} \cancel{r} = \cancel{r} = 1}$

// required for perf. tracking

$\boxed{y = r - \frac{H^\circ}{1+G^\circ K_{CL}} d}$

// nominal car model in feedb

trust the algebra

② let $K_{CL} = 10$,

$y = r - \frac{5}{1+20(10)} d \approx r - 0.025d$

1% uphill → −0.025 mph!

③ $y = \frac{1+G^\circ K_{fb}}{1+G K_{fb}} * \frac{G}{G^\circ} r$

$G/G^\circ = 1.1$ again → $1.0004r$

0.04% error in speed

Sensitivity Analysis

$\boxed{S_\theta(y)} = \%\Delta y / \%\Delta\theta = \frac{\Delta y / y}{\Delta\theta/\theta} = \frac{dy/y}{d\theta/\theta} = \boxed{\frac{dy}{d\theta} * \frac{\theta}{y}}$

subscript notation may flip!

- can be used as another way to find robustness: Ex $S_y(G) = \frac{\partial y}{\partial G} \frac{G}{y}$

MODELS

u → [M] → y

MIMO: a, b ⇒ [M] ⇒ x, y, z

cascade: u → [P] → v → [Q] → y

Q(P(u))

Delay Operator

u → [D_T] → y $y(t) = u(t-T)$

Memoryless

y(t) depends SOLELY on u(t)

if y(t) depends on any other values, it is dynamic.

$y(t) = u^2(t)$ ✓

$y(t) = SAT(u(t))$ ✓

$y(t) = u(t) + u(t-1)$ ✗

Causality

causal: $y(t) \to u(\tau)$ $\tau \le t$

strictly: $y(t) \to u(\tau)$ $\tau < t$

// All physical models are causal, → nothing depends on the future.

LINEARITY

① $M(\alpha u_1) = \alpha M(u_1)$

② $M(u_1 + u_2) = M(u_1) + M(u_2)$

TIME - INVARIANCE → typically, we deal with T-INV

$MD_\tau u = D_\tau M u$ for any u and Time-delay τ.

what?

⇒ "delayed input will result in delayed output of same expected form"

common types of applications

Cruise-control for cars

$m\dot{y} = Eu - mg\sin\theta - \alpha y$

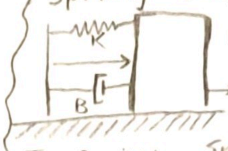
ma, engine, slope g, drag

Inverted Pendulum

$\frac{L}{3}\ddot{\theta} + \frac{1}{3}\ddot{x}\cos\theta = -\frac{1}{3}g\sin\theta$

- nonlinear
- time INVAR.

Spring - mass damper



$m\ddot{y} - B\dot{y} + Ky = u$

Linear (assuming spring linear)

Time-inv,

General Form

$y \to$ INPUT

$u \to$ OUTPUT

$y^{[n]} = f(y^{[n-1]}, y^{[n-2]}, \ldots \dot{y}, y, u^{[n-1]}, u^{[n-2]}, \ldots \dot{u}, u)$

nth-order

// sanity check

CAUSALITY HACK

degree of model $= n$ // highest derivative of y

relative degree $\delta = n - m$ // m is " " of u

- if ($\delta \geq 0$) → CAUSAL
- if ($\delta > 0$) → strictly causal

PARAMETER EST

① Parameter estimation - use physics to guess form

vs

② System ID - just say screw it and guess

Example: $m\dot{y} = Eu - \alpha d$ → given u^d, y^d

Trying to guess α: $y = M(u^d, \alpha)$

find norm of error $\| y^d - M(u^d, \alpha) \|$ for many α

$\min_\alpha \| y^d - M(u^d, \alpha) \|$

FIRST ORDER SYSTEMS. $\dot{y} + ay = bu$

General sol → $y(t) = \varepsilon e^{-at} + \int_0^t e^{-a(t-\tau)} bu(\tau)d\tau$

εe^{-at} → $\boxed{\text{free response}}$, due ONLY to initial condition $y(0) = \varepsilon$

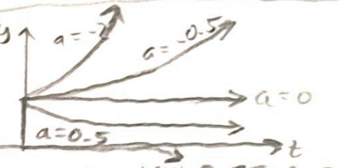
$\int_0^t e^{-a(t-\tau)} bu(\tau)d\tau$ → $\boxed{\text{Forced response}}$ (due only to input u and no init cond.)

Notice: εe^{-at} (free resp) when

$a < 0$ → y blows up

$a = 0$ → y stays const

$a > 0$ → $y \to 0$



$\frac{1}{a}$ → time const of decay for ASYMP STABLE

Time const. HACK:

t	T	2T	3T	4T
y(t)	0.37	0.14	0.05	0.02

Step response of F-O systems

Def: Step response → integrate forced resp $\int_0^u$ (u is stepsize)

Criteria ① $\varepsilon = 0$ (Free resp dies)

② Assume system stable

General Step-Resp: $y(t) = \underbrace{e^{-at}\left[y_0 - \frac{b}{a}u_{ss}\right]}_{\text{transient.}} + \underbrace{\frac{b}{a}u_{ss}}_{ss}$

Since we assumed stability, transient → 0

$y_{ss} = \frac{b}{a}u_{ss}$ → DC gain $= b/a$

Proportional Control

$u = K_p(r - y) + K_o r$

$= (K_p + K_o)r - K_p y$

$m\dot{y} + \alpha y = Eu$

how to choose K_p, K_o

$m\dot{y} + \alpha y = EK_p(r - y) + EK_o r$

when $y_{ss} = r$,

$m\dot{y}^{\to 0} + \alpha r = EK_o r$

$\boxed{K_o = \alpha/E}$

how about K_p

$m\dot{y} + (\alpha + EK_p)y = (EK_p + EK_o)r$

$\dot{y} = \underbrace{-\frac{(\alpha + EK_p)}{m}}_{a} y + \underbrace{\frac{(EK_p + EK_o)}{m}}_{b} r$

$\tau = \frac{1}{a}$

$\boxed{\tau = \frac{m}{(\alpha + EK_p)}}$

Summary of prop. control

K_o: Feed-Forward gain used for perfect tracking

K_p: Feed-back gain for τ adjustment

K_p can be adjusted toward a target τ!

SINUSOIDAL RESP

Time shift: $A\cos(\omega(t-10)+\theta)$ — 10 second R-shift

vs

Phase shift: $A\cos(\omega(t)+\theta)$ — shift phase by θ

consider $\dot{y} + ay = bu$

$u(t) \begin{cases} 0 & t = 0 \\ U\cos(\omega t) & t > 0 \end{cases}$

$y(t) = \underbrace{e^{-at}y_0 - \frac{Uab}{a^2+\omega^2}e^{-at}}_{\text{transient}} + \underbrace{\frac{Ub}{a^2+\omega^2}(\omega\sin(\omega t) + a\cos(\omega t))}_{\text{steady state}}$

→ $y_{ss} = M \cdot U\cos(\omega t + \phi)$

$M = \frac{|b|}{\sqrt{a^2+\omega^2}} = \left|\frac{b}{a + j\omega}\right|$ — amplification

$\phi = -\tan^{-1}\left(\frac{b}{a}\right) = \measuredangle\left(\frac{b}{a + j\omega}\right)$ — phase shift

// General Resp.

Bode Plots: $20\log_{10} M(\omega)$ vs $\log_{10}\omega$

$\phi(\omega)$ vs $\log_{10}\omega$

complex arithmetic

$|\frac{1}{j}| = 1$ $\measuredangle \frac{1}{j} = -90°$

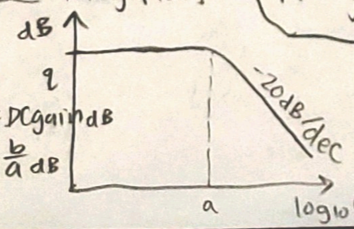
$e^{j\theta} = \cos\theta + j\sin\theta$

$\cos\theta = \frac{e^{j\theta} + e^{-j\theta}}{2}$

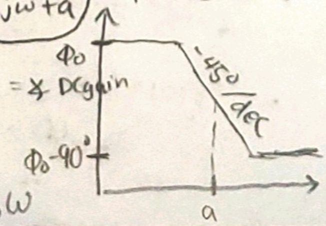
$\sin\theta = \frac{e^{j\theta} - e^{-j\theta}}{2}$

$H = \frac{b}{j\omega + a}$

Mag Plots



Phase Plots



Tips for hand-sketching Bode

Table: |corner f| {type {location

|p|/|z| Z/P LHP/RHP

magnitude:

① DC gain is the starting point

② ↑ slope 20dB/dec if it's a Z

↓ slope 20dB/dec if it's a P

③ poles of ↑ multiplicity n

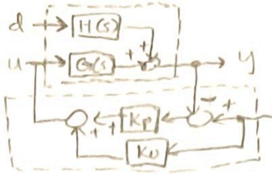
↕ slope by $20n$ dB/dec

Additional Midterm 2 material

Classical Control: Time Domain (99% of controllers out there)

$\boxed{u = K_p(r-y) + K_0 r}$ //controller

$\boxed{y = G(s)u + H(s)d}$ //plant (d is disturbance)



Let $G(s) = \frac{b}{s+a}$ $H(s) = \frac{h}{s+a}$

(AKA, this is a first-order plant)

$\dot{y} + ay = bu$ //plant

$u = K_p(r-y) + K_0 r$ //controller

$$\boxed{y = \left[\frac{bK_p + bK_0}{s+a+bK_p}\right] r + \left[\frac{h}{s+a+bK_p}\right] d}$$

Design of K_p, K_0 for certain criteria

① Perfect Tracking (DC gain = 1) $\boxed{K_0 = \frac{a}{b}}$

② Design of time const τ $\boxed{K_p = \frac{1}{b\tau} - \frac{a}{b}}$

Prop Control's pitfall: does not reject const disturbances! (DC gain of TF from d to y $\neq 1$).

Notice! → As $K_p \uparrow$, $\tau \downarrow$, $\frac{h}{s+a+bK_p} \downarrow$ ⟺ As $K_p \uparrow$, time constant ↓, disturbance reject ↑

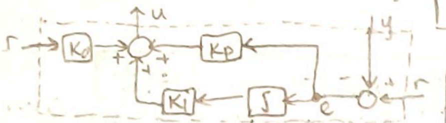
PIC

plant stays same, controller is different!

$y = \frac{b}{s+a}u + \frac{h}{s+a}d$

$\boxed{u = \left[K_p + \frac{K_i}{s}\right](r-y) + K_0 r}$

idea: - Perfect Tracking & Disturbance reject } satisfied by default!



$$\boxed{y = \left[\frac{(bK_p + bK_0)s + bK_i}{s^2 + (a+bK_p)s + bK_i}\right] r + \left[\frac{hs}{s^2 + (a+bK_p)s + bK_i}\right] d}$$

$\boxed{K_i = \frac{\omega_n^2}{b}}$

$\boxed{K_p = \frac{2\zeta\omega_n - a}{b}}$

//K_p and K_i can be used to determine damping and natural frequency (ω_n, ζ)

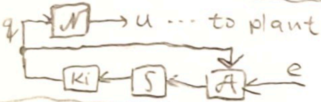
Proof of why Integral Works: if plant is stable, MUST converge to constant values. Thus ess (integral input) must → 0. $(r-y) \to 0$ ✓

Intuition: observes all PAST values of the error as well as current → desires to level out TOTAL error.

Bumpless Transfer (setting initial integral conditions) → $\boxed{z(t_0) = \frac{r}{G(0)K_i} - \frac{K_0 r}{K_i}}$

Where PID fails
① very nonlinear plants
② v high order plants
③ v high performance criteria
④ MIMO

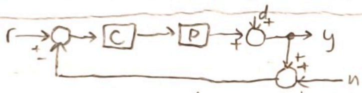
Actuator Saturation & Anti-windup



$\boxed{N}$: saturation block: does not allow signal to exceed $m < q < M$.

$\boxed{A}$: Anti-windup block (2-input 1-output): $f = \begin{cases} 0 & q(t) = M;\ e > 0 \\ 0 & q(t) = m;\ e < 0 \\ e & \text{otherwise} \end{cases}$

Classical Control: Freq. Domain



$y = \left[\frac{PC}{1+PC}\right] r + \left[\frac{1}{1+PC}\right] d - \left[\frac{PC}{1+PC}\right] n$

n vs d vs r → low freq reference; high freq noise; med freq disturbance

let $L = P^0 C$ { ① $\frac{L}{1+L} \approx 1$ @ freq for r $|L| \gg 1$; ② $\frac{1}{1+L} \approx 0$ @ freq for d $|L| \gg 1$; ③ $\frac{L}{1+L} \approx 0$ @ freq for n $|L| \ll 1$

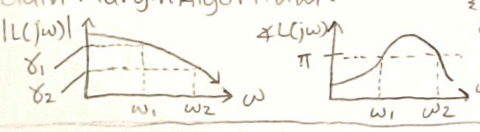
(A, B, C) are Poles; design C(s) to satisfy these poles.

Gain Margins

$\boxed{P_{true} = KP^0}$ K is true plant modelling error.

$K \in [\underline{K}, \bar{K}]$

Gain Margin Algorithm:

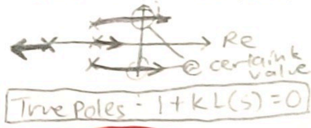


$\{\gamma_1, \gamma_2 \dots \gamma_n\}$ are the key crossover frequencies.

$\boxed{\begin{array}{l} \text{for all } \gamma_k > 1,\ K_{min} = \max\{\frac{1}{\gamma_k} \dots\} \text{ else} = 0 \\ \text{for all } \gamma_k < 1,\ K_{max} = \min\{\frac{1}{\gamma_k} \dots\} \text{ else} = \infty \end{array}}$

Recall: $y = \left[\frac{L}{1+L}\right] r$

closed-loop poles: roots of $1+L$.



$\boxed{\text{True Poles}: 1 + KL(s) = 0}$

Summary: $\boxed{0 < K_{min} < 1 < K_{max} < \infty}$

Time-Delay Margins:

$y = \left[\frac{L}{1+L}\right] r$ $1 + e^{-j\omega T} L^0(j\omega) = 0$ //set poles to 0

$L^0(j\omega) = -e^{-j\omega T}$ ⟨ $\boxed{|L^0(j\omega)| = 1}$, $\boxed{\measuredangle L^0(j\omega) = \pi - \omega T = \theta}$

Algorithm:
- find ω_k where $|L^0(j\omega)| = 1$ (these are gain-Xover freq).
- find θ_k @ ω_k → if $\theta_k > \pi$, $T_k = \frac{\theta_k - \pi}{\omega_k}$
- $T_{max} = \min(T_1, \dots T_k)$ → if no $\theta_k > \pi$, $T_{max} = \infty$

State-Space Analysis.

Problems with classical control
- cannot easily handle MIMO, nonlinear systems, v high order systems
- Does not systematically address robustness.

$\Sigma \sim \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array}\right]$ for SS realizations of block diagrams, always try to tie back to this form.

$\dot{x}(t) = Ax + Bu$
$y(t) = Cx + Du$

From Σ to tf $H(s)$

$\boxed{H(s) = C(sI - A)^{-1}B + D}$

Key property of A: the poles of tf H(s) are going to be a subset of the eigenvals of A

Ex: $y = x_1$ $\dot{y} = \dot{x}_1 = x_2$ $\ddot{y} = \ddot{x}_1 = \dot{x}_2 = x_3$

let $\dot{x}_1 = x_2$ $y = x_1$
$\dot{x}_2 = x_3$
$\dot{x}_3 = y^{[3]} = -7x_1 - 4x_2 - 3x_3 + u$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ y \end{bmatrix} = \underbrace{\left[\begin{array}{ccc|c} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -7 & -4 & -3 & 1 \\ \hline 1 & 0 & 0 & 0 \end{array}\right]}_{\Sigma} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ u \end{bmatrix}$$

Controllable Canonical Form

$$\Sigma = \left[\begin{array}{cccc|c} 0 & 1 & 0 & 0 \dots & 0 \\ & \ddots & & & \vdots \\ -\alpha_0 & \dots & & -\alpha_{n-1} & 1 \\ \hline \beta_0 & \dots & & \beta_{n-1} & d \end{array}\right] \Leftrightarrow \boxed{H(s) = \frac{\beta_{n-1}s^{n-1} + \dots + \beta_0}{s^n + \alpha_{n-1}s^{n-1} \dots \alpha_0} + d}$$

LinAlg Tips

① A is invertible ⟺ $\det \neq 0$, lin indep

② Block-partitioned matrix inverse $\begin{bmatrix} X & Y \\ 0 & Z \end{bmatrix}^{-1} = \begin{bmatrix} X^{-1} & -X^{-1}YZ^{-1} \\ 0 & Z^{-1} \end{bmatrix}$

③ $\mathrm{Spec}(A) = \{\lambda_1 \dots \lambda_n\}$

④ complex eigenvals/vectors come in conjugate pairs.

⑤ $(AB)^{-1} = B^{-1}A^{-1}$
$(AB)^T = B^T A^T$

⑥ Spectral Theorem: $\underset{n \times n \text{ simple}}{A} = T \Lambda T^{-1}$ $T = \begin{bmatrix} | & & | \\ v_1 & \cdots & v_n \\ | & & | \end{bmatrix}$ $\Lambda = \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix}$

⑦ Similar & Simple matrices.
Similar: $A \leftrightarrow T^{-1}AT$ //have the same eigenvalues
Simple: all eigenvals are real & distinct.

⑧ Polynomial evaluations of matrices
$P(x) =$ *some polynomial* $P(A) = T \begin{bmatrix} P(\lambda_1) & & 0 \\ & \ddots & \\ 0 & & P(\lambda_n) \end{bmatrix} T^{-1}$

$e^{At} = T \begin{bmatrix} e^{\lambda_1 t} & & 0 \\ & \ddots & \\ 0 & & e^{\lambda_n t} \end{bmatrix} T^{-1}$ $A^n = T \begin{bmatrix} \lambda_1^n & & 0 \\ & \ddots & \\ 0 & & \lambda_n^n \end{bmatrix} T^{-1}$

Tips for sketching phases:

1. find $\angle H(0)$ this is where phase plot starts.
2. VERY IMPORTANT
   - P → LHP → $-45°/dec$
   - P → RHP → $45°/dec$
   - Z → LHP → $45°/dec$
   - Z → RHP → $-45°/dec$

   all take effect in zone $(0.1p, 10p)$

Transfer function unique formats:

1. s-format

$a(s)y = b(s)u \Rightarrow \boxed{y = \frac{b(s)}{a(s)}u}$

$H(s) = b(s)/a(s)$

causal if: $\deg(a) \geq \deg(b)$

strictly causal if: $\deg(a) > \deg(b)$

Relative degree $= \deg(a) - \deg(b)$

2. Matrix format (MIMO): $\ddot{y}_1 + 2\dot{y}_1 + 6y_1 = \ddot{u}_1 + \dot{u}_2 - 3u_3$

$\dot{y}_2 + 3y_2 = 2u_2 - u_3$

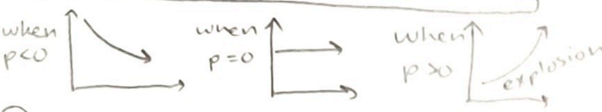
$[H(s)] = [A]_{2\times3}$ → 2 × ys → 3 × us

The TOTAL response: $y = H(s)u$

$y(t) =$ free resp + forced resp.

1. Free resp of DISTINCT REAL POLES

$$\boxed{y(t)_{free} = c_1e^{p_1t} + \dots + c_ne^{p_nt}}$$



2. Free resp of DISTINCT COMPLEX POLES $(\sigma \pm j\omega)$

$$\boxed{y(t)_{free} = e^{\sigma t}[c_1\sin(\omega t) + c_2\cos(\omega t)]}$$

- oscillates
- when $\sigma < 0$, resp $\xrightarrow{appr}$ 0
- when $\sigma > 0$, resp $\xrightarrow{apprs} \infty$
- when $\sigma \uparrow$, more drastic change
- when $\omega \uparrow$, oscillation freq ↑

3. Repeated Poles.

Real: $p \times n \rightarrow \boxed{y_{free}(t) = c_1e^{p_1t} + tc_2e^{p_1t} + \dots + t^{n-1}c_ne^{p_1t}}$

Complex: $\sigma \pm j\omega \times n \rightarrow \boxed{y_{free}(t) = e^{\sigma t}[c_1\sin(\omega t) + c_2\cos(\omega t)] + te^{\sigma t}[\text{" "}] + \dots + t^{n-1}e^{\sigma t}[\text{" "}]}$

Random Aside

$D_T \Rightarrow \boxed{H(s) = e^{-sT}}$ $|H(s)| = 1$ $\angle H(s) = -\omega T$

$\int \Rightarrow \boxed{\frac{1}{s}}$ $u(\frac{1}{s}) = y$ $\dot{y} = u$ $y = \int u\,dt$

Routh Test:

s^n	a_0	a_2	$a_4 \dots$
s^{n-1}	a_1	a_3	$a_5 \dots$
s^{n-2}	b_1	b_2	$b_3 \dots$
s^{n-3}	c_1	c_2	$\dots$
$\vdots$			
s^1	e_1		
s^0	f_1		

$b_1 = \frac{a_1a_2 - a_0a_3}{a_1}$

$b_2 = \frac{a_1a_4 - a_0a_5}{a_1}$

$b_3 = \frac{a_1a_6 - a_0a_7}{a_1}$

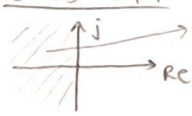
$c_1 = \frac{b_1a_3 - a_1b_2}{b_1}$

$c_2 = \frac{b_1a_5 - a_1b_3}{b_1}$

Remember to always divide by corner values

# of sign changes in this critical array = # of poles in RHP = 0 for stability.

STABILITY



Re{poles} < 0 !

All poles in LHP for stable systems

For 2nd order TF,

$H(s) = */(as^2 + bs + c)$

if no sign changes between a, b, c, stable!

Sinusoidal Response vs Step Response

$y = H(s)u$ where $H(s)$ is stable

$u = U\cos(\omega t + \phi)$

$$\boxed{y_{ss} = MU\cos(\omega t + \phi + \theta) + y_{transient}}$$

$\boxed{M = |H(j\omega)|}$ $\boxed{\theta = \angle H(j\omega)}$

$y = H(s)u$ $u = \begin{cases} U_{ss} & t \geq 0 \\ 0 & t < 0 \end{cases}$

$y_{ss} = H(0)U_{ss} + y_{trans}$

$H(0)$ → DC-gain.

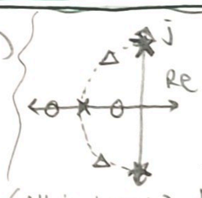
Complex Math (Random)

$\boxed{e^{j\pi} = -1}$ $\boxed{e^{j\phi} = \cos(\phi) + j\sin(\phi)}$

2nd Order Systems $y = H(s)u$ $H(s) = (b_2s^2 + b_1s + b_0)/(a_2s^2 + a_1s + a_0)$

simple form $\boxed{\frac{K}{s^2 + 2\varepsilon\omega_ns + \omega_n^2} = H(s)}$

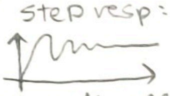
- $\omega_n > 0$ nat freq
- $\varepsilon > 0$ damping
- DC gain (when $K = \omega_n$) = 1



(All in LHP!)

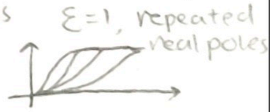
△ → Underdamped 2 complex poles

step resp:



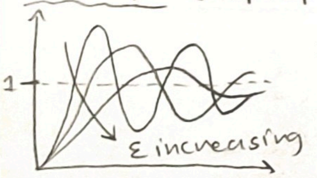
✱ - undamped purely complex poles

✕ → critically damped $\varepsilon = 1$, repeated real poles

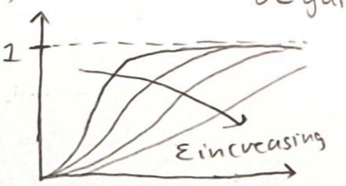


O → overdamped $\varepsilon > 1$ - 2 real poles

ε vs output (step resp)

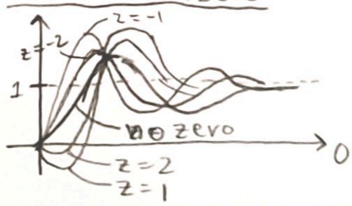


UNDERDAMPED

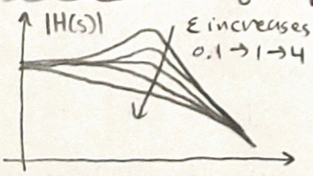


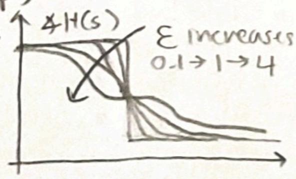
OVERDAMPED

Effect of a zero

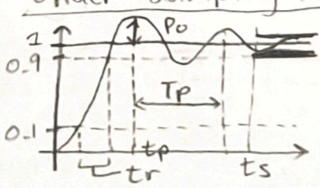


ε vs output (freq resp)





Under-damping features



t_r = 10-90% RISE time $\approx \boxed{1.8/\omega_n}$

t_{pe} = Peak time $= \boxed{\frac{\pi}{\omega_n\sqrt{1-\varepsilon^2}}} = \boxed{\pi/\omega_d}$

t_s = 95% settling time $= \boxed{3/\varepsilon\omega_n}$

PO = Percent overshoot $= \boxed{100 * e^{-\pi\varepsilon/\sqrt{1-\varepsilon^2}}}$

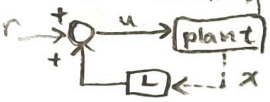
ω_d is oscillation freq

T_p = oscillation period $= \boxed{\frac{2\pi}{\omega_d}}$

state space Design

① State FeedBack for movement of eigen values

Why unhappy w/ current e-vals of A?
- unstable - need different damping
- slow settling - magnitude not match.



$\boxed{u = Lx + r}$

L: m×n matrix to be designed
m → # of inputs
n → # of states

$\Sigma \begin{cases} \dot{x} = Ax + Bu \\ y = Cx \end{cases}$ becomes $\Sigma_{CL} \begin{cases} \dot{x} = [A + BL]x + Br \\ y = Cx \end{cases}$

$\boxed{A_{CL} = A + BL}$

Ex: $\underbrace{\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ a & b & c \end{bmatrix}}_{\text{A (controllable canon)}} + \underbrace{\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}}_{B} \underbrace{[l_1 \ l_2 \ l_3]}_{\text{L (design)}} = \underbrace{\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ a+l_1 & b+l_2 & c+l_3 \end{bmatrix}}_{A_{CL} \text{ (result)}}$

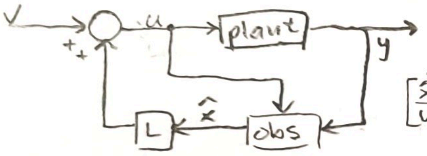
Algorithm
- start with Σ
- change states $\hat{x} = T^{-1}x$ to get CCF
- $u = \hat{L}\hat{x} = \hat{L}T^{-1}x$

OR

- start with $\Sigma(A,B,C,D)$
- decide desired poles $(p_1, p_2, p_3, \dots)$
- set $\det(sI - A_{CL}) = (s-p_1) \times \cdots \times (s-p_n)$
- solve for L.

combine the BEST of both worlds

output Feedback Stabilization

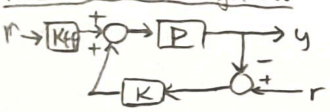


$\boxed{\dot{\hat{x}} = (A+KC)\hat{x} + BL\hat{x} + K*(r-y)}$

$\begin{bmatrix} \dot{\hat{x}} \\ u \end{bmatrix} = \boxed{\left[\begin{array}{c|c} A+BL+KC & K \\ \hline L & 0 \end{array}\right]} \begin{bmatrix} \hat{x} \\ r-y \end{bmatrix}$

The FINAL controller

Feed-forward gain.



$y = \left[\dfrac{P(s)K(s) + P(s)K_{ff}}{1 + P(s)K(s)}\right] r$

for perfect tracking, set DC gain = 1

$\dfrac{P(0)K(0) + P(0)K_{ff}}{1 + P(0)K(0)} = 1 \Rightarrow P(0)K_{ff} = 1$

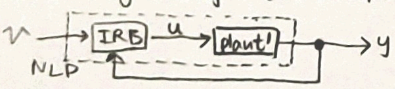
$\boxed{K_{ff} = 1/P(0) = 1/C(-A)^{-1}B}$

what if A is non-invertible? (thought experiment)... integral already there

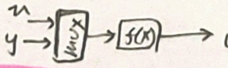
TACTIC 1: Feedback Linearization.

Ex: magnetically suspended ball: $\ddot{y} = g - \frac{c}{m}\frac{u^2}{y^2}$

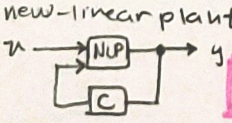
set $v = g - \frac{c}{m}\frac{u^2}{y^2}$ (new input) $\Rightarrow \ddot{y} = v$



IRB: "input refining box" $u = y\sqrt{(g-v)m/c}$ ★ needs y!



NLP: new-linear plant.



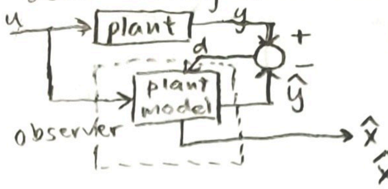
$\boxed{\text{LINEARIZED BY FEEDBACK.}}$

TACTIC # 2: Equilibrium points

Definition ★ (x_{eq}, u_{eq}, y_{eq})
- if we let $x(0) = x_{eq}$
- set input to constant $u = u_{eq}$
  - $x(t) \xrightarrow{\text{stays}} x_{eq}$
  - $u(t) \xrightarrow{\text{stays}} u_{eq}$
  - $y(t) \xrightarrow{\text{stays}} y_{eq}$

② Observers → A PIECE OF SOFTWARE!

used during a "sensor shortage"



$\boxed{d = y - \hat{y}}$ ("discrepancy") represents error btw true plant output & model plant output

$\hat{x}$: observer-predicted states ★

Algebra: plant model within observer:

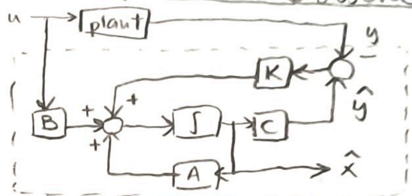
$\dot{x} = Ax + Bu$ → $\dot{\hat{x}} = A\hat{x} + Bu + K(\hat{y} - y)$ correction factor
$y = Cx$ → $\hat{y} = C\hat{x}$

$\hat{x}$ → output of observer.

$\boxed{\dot{\hat{x}} = (A+KC)\hat{x} + Bu - Ky}$

obs ss realization ~ $\left[\begin{array}{c|c|c} A+KC & B & -K \\ \hline I & 0 & 0 \end{array}\right]$
- inputs noisy plant output y and u
- outputs state estimate $\hat{x}$

Block diagram inside observer



Dynamics of Observer

let $e = \hat{x} - x$

$\dot{e} = \dot{\hat{x}} - \dot{x}$
$= (A+KC)e$ $\qquad e(0) = \varepsilon$

$\boxed{e(t) = e^{(A+KC)t}\varepsilon}$

critical for consid when designing K ← where error decay rate = eig (A+KC)

Kalman filter theory → ideal observer
- rely more on model if real y is noisy (sensors suck)
- rely more on real measurements if plant model sucks.
- correction factor K → K↑ : sensor is good; K↓ : plant model is good

Nonlinear systems

General form: $y^{[n]} = f(y^{[n-1]}, \dots, \dot{y}, y, u^{[m]}, \dots, \dot{u}, u)$
where f is some complicated nonlinear function

ss-realization of nonlinear systems

Ex: $\ddot{y} + \underbrace{3\dot{y}y}_{\text{all}} + \underbrace{7\cos(y)}_{\text{non}} = \underbrace{4u^2}_{\text{lin}} + \underbrace{uy}_{\text{ear}}$

$f_1, f_2 \dots h$ are ANY function

$x_1 = y \quad \dot{x}_1 = \dot{y}$
$x_2 = \dot{y} \quad \dot{x}_2 = \ddot{y}$

$\dot{x}_1 = f_1(x_1, x_2, u) = x_2$
$\dot{x}_2 = f_2(x_1, x_2, u) = -3x_1x_2 - 7\cos x_1 + 4u^2 + ux_1$
$y = h(x_1, x_2, u) = x_1$

General ss realization

$$\boxed{\begin{aligned} \dot{x}_1 &= f_1(x_1, \dots, x_n, u_1, \dots u_m) \\ \dot{x}_n &= f_n(x_1, \dots, x_n, u_1, \dots u_m) \\ y_1 &= h_1(x_1, \dots, x_n, u_1, \dots u_m) \\ y_p &= h_p(x_1, \dots, x_n, u_1, \dots u_m) \end{aligned}} \Rightarrow \begin{aligned} \dot{x} &= f(x,u) \to \text{state dynamics} \\ y &= h(x,u) \to \text{output} \end{aligned}$$

f & h are vector-valued functions

$\dot{x}$ and y can be vectors

Algorithm

$\dot{x} = f(x,u) \quad \dot{x}_{eq} = 0$
$y = h(x,u)$

① set $f(x,u) = 0$ (all derivatives of states = 0)
② find all $(x,u)_i$ that satisfy
③ for each $(x,u)_i$, set $y_{eq\,i} = h(x,u)_{eq\,i}$

Parameterization (by $u_{eq} = \lambda$ in this example)

$\dot{x}_1 = f_1(x_1, x_2, u) = x_2$
$\dot{x}_2 = f_2(x_1, x_2, u) = 10 - \frac{u^2}{15x_1^2}$
$y = h(x_1, x_2, u) = x_1$

$\left.\begin{aligned} x_{1eq} &= 0.0816\,u_{eq} \\ x_{2eq} &= 0 \\ y_{eq} &= x_{1eq} \end{aligned}\right\}$ continuum

set $u_{eq} = \lambda$: $(x_{eq}, u_{eq}, y_{eq}) = \left(\begin{bmatrix} 0.0816\lambda \\ 0 \end{bmatrix}, 0.0816\lambda, \lambda\right)$

JACOBIAN LINEARIZATION

Jacobian Review:

scalar function Jacobian: $\phi(x_1, x_2, \dots x_n) \rightarrow \boxed{\nabla\phi = \left[\frac{\partial\phi}{\partial x_1}, \frac{\partial\phi}{\partial x_2}, \dots, \frac{\partial\phi}{\partial x_n}\right]}$ scalar function yields a vector - Jacobian

vector-valued Jacobian: $\vec{\phi}(\vec{x}) = \vec{\phi}(x_1, x_2, \dots x_n) = \begin{bmatrix} \phi_1(x_1, x_2, \dots x_n) \\ \vdots \\ \phi_m(x_1, x_2, x_n) \end{bmatrix}$ $\boxed{\nabla\phi = \begin{bmatrix} \frac{\partial\phi_1}{\partial x_1} & \cdots & \frac{\partial\phi_1}{x_n} \\ \vdots & \ddots & \\ \frac{\partial\phi_m}{x_1} & \cdots & \frac{\partial\phi_m}{x_n} \end{bmatrix}}$ $m \times n$ matrix

vector function yields a MATRIX Jacobian.

Intuition behind the Jacobian

Background: Taylor estimation

$$\phi(x) \approx \phi(\varepsilon) + \boxed{\left.\frac{d\phi}{dx}\right|_{x=\varepsilon}} * \underbrace{(x-\varepsilon)}_{\text{works when this value is small}}$$

Jacobian Acts as placeholder for derivative when dealing w/ vector-valued functions.

for a NL function $\vec{\phi}(x_1, x_2)$ $\boxed{\phi(x) \approx \phi(\varepsilon) + \nabla\phi(\varepsilon) * (x-\varepsilon)}$

// Row 'n' of the Jacobian is the sum of all partial derivatives of $\phi_n(x)$ for all $x_1, x_2 \dots x_m$!

How to employ the Jacobian

★ $\varepsilon = (x_{eq}, u_{eq}, y_{eq})$

① $\left.\begin{aligned} \delta x &= x - x_{eq} \\ \delta u &= u - u_{eq} \\ \delta y &= y - y_{eq} \end{aligned}\right\}$ simple definitions

② Linearize $f(x,u) = \dot{x}$

$\delta\dot{x} = \dot{x} - \cancel{\dot{x}_{eq}}^{\,0} = \dot{x} = f(x,u)$

$\approx \cancel{f(x_{eq}, u_{eq})} + \nabla f(\varepsilon)\left(\begin{bmatrix} x \\ u \end{bmatrix} - \begin{bmatrix} x_{eq} \\ u_{eq} \end{bmatrix}\right)$

0 (because it is an eq. pt.

$\Rightarrow \delta\dot{x} = \nabla f(\varepsilon)\left(\begin{bmatrix} x \\ u \end{bmatrix} - \begin{bmatrix} x_{eq} \\ u_{eq} \end{bmatrix}\right)$

$$= \underbrace{\begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_n}{x_1} & \cdots & \frac{\partial f_n}{x_n} \end{bmatrix}}_{A\,(n \times n)} \underbrace{(\overrightarrow{x - x_{eq}})}_{(n \times 1)} + \underbrace{\begin{bmatrix} \frac{\partial f_1}{\partial u_1} & \cdots & \frac{\partial f_1}{\partial u_m} \\ \vdots & \ddots & \\ \frac{\partial f_n}{\partial u_1} & \cdots & \frac{\partial f_n}{\partial u_m} \end{bmatrix}}_{B\,(n \times m)} \underbrace{(\overrightarrow{u - u_{eq}})}_{(m \times 1)}$$

$\delta\dot{x} = A(\delta\vec{x}) + B(\delta\vec{u})$

③ Linearize $h(x,u)$

$\delta y = y - y_{eq}$

$= h(x,u) - y_{eq}$

$= \cancel{h(x_{eq}, u_{eq})} + \nabla h(\varepsilon)\left(\begin{bmatrix} x \\ u \end{bmatrix} - \begin{bmatrix} x_{eq} \\ u_{eq} \end{bmatrix}\right)$

$- \cancel{y_{eq}}$ // $h(x_{eq}, u_{eq}) - y_{eq} = 0$!

$\delta y = \nabla h(\varepsilon)\left(\begin{bmatrix} x \\ u \end{bmatrix} - \begin{bmatrix} x_{eq} \\ u_{eq} \end{bmatrix}\right)$

$$= \underbrace{\begin{bmatrix} \frac{\partial h_1}{\partial x_1} & \cdots & \frac{\partial h_1}{\partial x_n} \\ & \ddots & \\ \frac{\partial h_p}{\partial x_1} & \cdots & \frac{\partial h_p}{\partial x_n} \end{bmatrix}}_{C\,(p \times n)} \underbrace{[\overrightarrow{x - x_{eq}}]}_{(n \times 1)} + \underbrace{\begin{bmatrix} \frac{\partial h_1}{\partial u_1} & \cdots & \frac{\partial h_1}{\partial u_m} \\ \vdots & \ddots & \\ \frac{\partial h_p}{\partial u_1} & \cdots & \frac{\partial h_p}{\partial u_m} \end{bmatrix}}_{D\,(p \times m)} \underbrace{[\overrightarrow{u - u_{eq}}]}_{(m \times 1)}$$

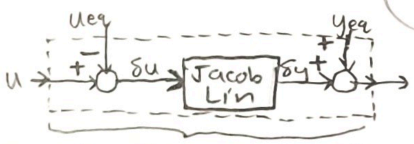
$\delta y = C(\delta\vec{x}) + D(\delta\vec{u})$

SYSTEM

$$\boxed{\begin{aligned} \delta\dot{x} &= A\delta x + B\delta u \\ \delta y &= C\delta x + D\delta u \end{aligned}}$$

$$\begin{bmatrix} \delta\dot{x} \\ \delta y \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} \delta x \\ \delta u \end{bmatrix}$$

// look familiar?



Nonlinear plant.